INTERVIEW CHEAT SHEET

[SELF INTRODUCTION]

Abhay Vishwakarma, penultimate-year B.Tech CS (AI & ML) from AKTU, proficient in Python, AI, ML, DSA, backend, and prompt engineering, aiming to secure a Software Engineering role that blends AI innovation with scalable systems.

[PROJECT PITCHES]

-> Multi-Agent AI Research System

Build by orchestrating specialized agents (Research, Scraper, Writer, Critic, LLM) with LangChain and Streamlit; difficulty in synchronizing agent workflows, solved with event-driven architecture and real-time visualization.

-> AI-Powered Telegram Bot

Constructed using Aiogram, Groq LLM, and async queues; challenge was low-latency concurrency, overcome via asynchronous message handling and connection pooling.

-> Movie Recommendation System

Developed dual-mode (content + collaborative) engine in Python with NumPy; struggled with large matrix operations, resolved by optimized vectorized calculations.

-> Context-Aware QA Chatbot

Implemented with LangChain, Chroma, and Groq LLM; faced prompt-latency trade-offs, addressed by efficient prompt engineering and caching.

-> Mini GPT Model

Built a decoder-only Transformer from scratch in PyTorch; difficulty in implementing masked self-attention and training stability, fixed with careful layer normalization and learning-rate scheduling.

[KEY TECHNOLOGIES]

Python	high-level language for rapid prototyping and data science.
C	systems programming language for low-level performance.
SQL	language for querying and managing relational databases.
Data Structures & Algorithms	core CS fundamentals for efficient problem solving.
OOP	paradigm organizing code into objects and classes.
DBMS	systems managing persistent data storage.
OS	software managing hardware resources and processes.
Computer Networks	study of communication protocols and data transfer.
Software Engineering	disciplined approach to building reliable software.
Multithreading	concurrent execution of threads within a process.
Async I/O	non-blocking input/output for scalable I/O operations.
Concurrency Patterns	reusable designs for safe concurrent execution.
FAST APIs	lightweight framework for building high-performance APIs.

Rate Limiting	controlling request frequency to protect resources.
Connection Pooling	reusing database connections to reduce latency.
Docker	container platform for consistent deployments.
Machine Learning	algorithms that learn patterns from data.
Deep Learning	neural networks with multiple layers for complex tasks.
Generative AI	models that produce new content (text, images, etc.).
LLMs	large language models providing contextual understanding.
NLP	techniques for processing human language.
RAG Pipelines	combining retrieval with generation for accurate answers.
LangChain	framework for building LLM-powered applications.
Streamlit	rapid web app development for data science.
PyCharm	integrated development environment for Python.
Aiogram	Telegram bot framework using asyncio.
Flask	micro web framework for lightweight APIs.
Pandas	data manipulation and analysis library.
NumPy	numerical computing with efficient arrays.
Git	distributed version control system.
GitHub	hosting platform for code repositories.
Chroma	open-source vector database for similarity search.
Pinecone	managed vector database for scalable similarity queries.

[DSA TOPICS TO REVISE]

• **Sliding Window**	technique for optimizing range queries over arrays.
• **Binary Search**	divide-and-conquer method for sorted data lookup.
• **Graphs**	structures of nodes and edges for connectivity problems.
• **Dynamic Programming**	solving problems by breaking them into overlapping subproblems.
• **String Algorithms**	methods for pattern matching and manipulation.

[HR ANSWERS]

Q: Why this role	I want to apply my AI and systems expertise to build scalable, AI-driven products that solve real business problems.
Q: Strength	Strong foundation in both AI and backend engineering, proven by production-ready projects and 300+ LeetCode solves.
Q: Weakness	Occasionally over-engineer solutions; I'm learning to balance simplicity with robustness.

Q: Long term goal	Lead an AI/ML product team focused on building end-to-end intelligent systems.
Q: Why you hired in this company	Your emphasis on cutting-edge AI and scalable engineering aligns with my skill set and career aspirations.
Q: What is your salary expectation	Open to discussion; I'm looking for a competitive package that reflects my experience and the role's responsibilities.

[WEAK AREAS TO COVER]

>> Advanced concurrency patterns for high-throughput services.

>> Database indexing and query optimization.

>> Security best practices for AI deployments.